A priori estimates for the wave equation with a self-adjoint operator squeezed between two Laplacians

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Editorial note. All delivered mathematical text is the author's own, copied from the frozen source sources/jep-2024-259-a-spectral-ansatz-for-the-long-time-homogenization/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The unit has no row in the census, so the gate is the only authority for this card, printed numbers are computed by hand and the finished PDF is verified with pdftotext (duty desk rule for this band).

Editorial note. The counted span carries no \cite and no reference to a label outside the span.

Editorial scope. Single Lemma card cut from doi:10.5802/jep.259 (A spectral ansatz for the long-time homogenization of the wave equation); statement at source lines 2383-2397 and the author's own proof at source lines 2398-2414.

Allowed standard prerequisites (editorial).

- The two-sided bound on the operator. The hypothesis of the statement, source line 2384. It makes the square root of the operator a bounded operator from the space into the homogeneous order minus one space with bounded inverse, which is what allows the divergence in the source term to be absorbed.
- Duhamel's formula for the wave equation. The representation used in the proof at source lines 2401-2406. It writes the solution as an integral of the sine propagator against the source terms, which is then estimated by spectral calculus.

Lemma 1.1 (A priori estimates)**.** *Let $\mathcal{L}$ be a self-adjoint operator on $\mathrm{L}^2(\mathbb{R}^d)$ satisfying the bound $-\triangle \le \mathcal{L} \le -C_0\triangle$ for some constant $C_0 < \infty$. Given $F_1 \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^+;\mathrm{L}^2(\mathbb{R}^d))$ and $F_2, F_3 \in W^{1,1}_{\mathrm{loc}}(\mathbb{R}^+;\mathrm{L}^2(\mathbb{R}^d))$ with $F_2|_{t=0} = F_3|_{t=0} = 0$, let z be the solution of the wave equation*

$$\begin{cases} (\partial_t^2 + \mathcal{L})z = F_1 + \nabla \cdot F_2 + \partial_t F_3 \quad \text{in } \mathbb{R}^d, \\ z|_{t=0} = \partial_t z|_{t=0} = 0. \end{cases}$$

Then, for all $t \geq 0$, we have

$$\|z^t\|_{\mathrm{L}^2(\mathbb{R}^d)} \lesssim_{C_0} t\|F_1\|_{\mathrm{L}^1((0,t);\mathrm{L}^2(\mathbb{R}^d))} + \|(F_2, F_3)\|_{\mathrm{L}^1((0,t);\mathrm{L}^2(\mathbb{R}^d))},$$

and

$$\|Dz^t\|_{\mathrm{L}^2(\mathbb{R}^d)} \lesssim_{C_0} \|(F_1, \partial_t F_2, \partial_t F_3)\|_{\mathrm{L}^1((0,t);\mathrm{L}^2(\mathbb{R}^d))}. \qquad \Diamond$$

Proof. The assumption $-\triangle \leq \mathcal{L} \leq -C_0\triangle$ entails that $\sqrt{\mathcal{L}}$ defines a bounded linear operator $\mathrm{L}^2(\mathbb{R}^d) \to \dot{H}^{-1}(\mathbb{R}^d)$ with bounded inverse. We may therefore define $\tilde{F}_2$ as the solution of $\sqrt{\mathcal{L}}\tilde{F}_2 = \nabla \cdot F_2$, which satisfies $\|\tilde{F}_2\|_{\mathrm{L}^2(\mathbb{R}^d)} \lesssim_{C_0} \|F_2\|_{\mathrm{L}^2(\mathbb{R}^d)}$. The solution z of the wave equation can then be represented in terms of Duhamel's formula,

$$z^t = \int_0^t \frac{\sin((t-s)\sqrt{\mathcal{L}})}{\sqrt{\mathcal{L}}} \, (F_1^s + \partial_s F_3^s) \, ds + \int_0^t \sin((t-s)\sqrt{\mathcal{L}}) \, \tilde{F}_2^s \, ds.$$

Integrating by parts in the integral for F_3, with $F_3|_{t=0} = 0$, this can be rewritten as

$$z^t = \int_0^t \frac{\sin((t-s)\sqrt{\mathcal{L}})}{\sqrt{\mathcal{L}}} \, F_1^s \, ds + \int_0^t \sin((t-s)\sqrt{\mathcal{L}}) \, \tilde{F}_2^s \, ds + \int_0^t \cos((t-s)\sqrt{\mathcal{L}}) \, F_3^s \, ds,$$

and the stated L^2-estimate follows from spectral calculus. For the energy estimate, we rather use energy conservation in form of

$$\begin{aligned}
\tfrac{1}{2}\partial_t \int_{\mathbb{R}^d} \big(|\partial_t z|^2 + z\mathcal{L}z + 2F_2 \cdot \nabla z\big) &= \int_{\mathbb{R}^d} (\partial_t z) \, (\partial_t^2 + \mathcal{L})z + \partial_t \int_{\mathbb{R}^d} F_2 \cdot \nabla z \\
&= \int_{\mathbb{R}^d} (\partial_t z)\big(F_1 + \partial_t F_3 + \nabla \cdot F_2\big) + \partial_t \int_{\mathbb{R}^d} F_2 \cdot \nabla z \\
&= \int_{\mathbb{R}^d} (\partial_t z)\big(F_1 + \partial_t F_3\big) + \int_{\mathbb{R}^d} \partial_t F_2 \cdot \nabla z \\
&\leq \|Dz\|_{\mathrm{L}^2(\mathbb{R}^d)} \|(F_1, \partial_t F_2, \partial_t F_3)\|_{\mathrm{L}^2(\mathbb{R}^d)},
\end{aligned}$$

and the conclusion follows. □